

MATHEMATICAL LITERACY

PAPER 1

STUDENT STUDY GUIDE

Meet Nova — Your Maths Literacy Mentor

Nova comes from the Latin for "new," like a star that suddenly shines bright. That's exactly his vibe — turning confusion into those "ohhh" moments. He's sharp, fast, and loves solving problems. Maths becomes patterns instead of panic. Accounting starts to feel structured instead of stressful. Nova shows you there's always a method behind the madness. When you're stuck, he finds a new way to look at it.



WHAT'S INSIDE

- P1** What the Paper Looks Like
 - P2** The 5 Topics That Matter Most
 - P3** Step-by-Step Methods with Worked Examples
 - P4** Mistakes That Cost Students Marks
 - P5** Your 14-Day Study Plan
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PART 1

What Does the Paper Look Like?

Maths Literacy Paper 1 tests basic skills. It is not random. The same topics appear every year. There are 5 questions. The paper is 150 marks in 3 hours. Most marks come from calculations, tables, percentages, and finance. Know the structure and you know where to score.

Q	What it Tests	Marks	Difficulty
Q1	Basic calculations, tables, ratios, simple percentages	~30	Easy
Q2	Finance: VAT, discounts, tax, bills, bank charges	~30	Easy-Med
Q3	Data handling: mean, median, range, graphs	~30	Medium
Q4	Measurement: area, volume, scale, maps, plans	~30	Medium
Q5	Mixed: finance, conversions, interest, percentage change	~30	Med-Hard

● **Tip:** 150 marks in 3 hours = **1.2 minutes per mark**. Q1 should take 30 minutes. Q2 finance gets 35 minutes. Easy marks live in tables and basic percentages. If stuck for more than 3 minutes, skip and come back.

Where the easy marks are: reading tables, finding totals, converting units, basic percentage calculations. Always show your formula, substitution, and units. You earn marks for method, not just the final answer.

Where students lose time: long word problems, scale conversions, multi-step finance questions. Read the question twice. Underline key numbers. Write the formula before you calculate.

PART 2

The 5 Topics That Give You the Most Marks

These 5 topics appear in **every single paper**. They make up over 120 marks out of 150. Master them and you are already there.

1 Finance: VAT, Interest, Salary

~35
marks

Q2

- VAT (15%): Price $\times$ 1.15 to add VAT. Price $\div$ 1.15 to remove VAT.
- Discount: New price = Original $\times$ (1 – discount %)
- Simple interest: $A = P(1 + i \times n)$
- Compound interest: $A = P(1 + i)^n$
- Salary: Gross – Deductions = Net

► **Strategy:** Always check if VAT is "included" or "excluded." Read every word.

2 Measurement: Area, Volume, Conversions

~25
marks

Q4

- Area of rectangle: $l \times b$. Area of circle: πr^2 .
- Volume of box: $l \times b \times h$. Volume of cylinder: $\pi r^2 h$.
- 1 m = 100 cm = 1 000 mm. 1 km = 1 000 m.
- 1 L = 1 000 mL. 1 kg = 1 000 g.
- 1 m² = 10 000 cm². 1 m³ = 1 000 L.

► **Strategy:** Convert all values to the same unit before you calculate.

3 Maps & Scale

~15
marks

Q4

- Scale 1 : 100 means 1 cm on map = 100 cm in real life.
- Real distance = Map distance $\times$ scale factor
- Map distance = Real distance $\div$ scale factor
- Always check the scale unit (cm, m, km).

► **Strategy:** Multiply for real life. Divide for map. Check units twice.

4 Data Handling: Tables & Graphs

~30
marks

Q3

- Mean = Sum of values $\div$ Number of values
- Median = Middle value (sort the list first)
- Mode = Value that appears most
- Range = Highest – Lowest
- Read graph axes and scale carefully.

► **Strategy:** Sort numbers in order before finding the median.

5 Probability & Percentages

~15
marks

Q1,
Q3

- Probability = Favourable outcomes $\div$ Total outcomes
- Answer can be a fraction, decimal, or percentage.
- Percentage change = $(\text{New} - \text{Old}) \div \text{Old} \times 100$
- Probability is always between 0 and 1.

► **Strategy:** Read the question. Answer in the format asked.

PART 3

How to Solve Each Question Type

Follow these steps every time. Each method has 3 worked examples — Easy, Medium, and Hard. Show formula. Substitute. Calculate. Add units.

METHOD 1

Finance Calculations

Step 1 — Read the question carefully

Find the price, rate, time, and what you must calculate. Underline key numbers.

Step 2 — Choose the correct formula

VAT: $\times 1.15$ (add) or $\div 1.15$ (remove). Simple interest: $A = P(1 + in)$. Compound: $A = P(1 + i)^n$.

Step 3 — Substitute values

Write the formula. Replace letters with numbers. Show the substitution.

Step 4 — Calculate and round

Use the calculator. Round only at the end. Add the unit (R, %, kg).

● **Tip:** VAT in South Africa is 15%. To add VAT, multiply by 1.15. To remove VAT, divide by 1.15.

WORKED EXAMPLE A

Easy

A jacket costs R450 before VAT. Calculate the price after 15% VAT is added.

Step 1 Price before VAT = R450.

Step 2 Formula: New price = Old $\times 1.15$

Step 3 = 450×1.15

Step 4 = R517.50

✓ **Price including VAT = R517.50**

WORKED EXAMPLE B

Medium

Thabo invests R5 000 at 8% simple interest per year. Calculate the value after 3 years.

Step 1 $P = \text{R5 000}$, $i = 0.08$, $n = 3$.

Step 2 Formula: $A = P(1 + in)$

Step 3 $A = 5\,000(1 + 0.08 \times 3)$

Step 4 $A = 5\,000(1.24) = \text{R6 200}$

✓ **Value after 3 years = R6 200**

WORKED EXAMPLE C

Hard

Lerato earns R12 500 per month. UIF is 1%. Tax is R900. Medical aid is R650. Calculate her net salary.

Step 1 Gross salary = R12 500.

Step 2 UIF = $1\% \times 12\,500 = R125$

Step 3 Total deductions = $125 + 900 + 650 = R1\,675$

Step 4 Net = $12\,500 - 1\,675 = R10\,825$

✓ **Net salary = R10 825**

Try It Yourself

A TV costs R8 050 (VAT included). Calculate the price before VAT.

Answer: $8\,050 \div 1.15 = R7\,000$

METHOD 2

Measurement & Conversions

Step 1 — Identify the shape and units

Rectangle, circle, box, cylinder. Note all units used in the question.

Step 2 — Convert to the same unit

Convert mm $\rightarrow$ cm or cm $\rightarrow$ m before calculating. Mixed units cause wrong answers.

Step 3 — Use the correct formula

Area: $l \times b$ or πr^2 . Volume: $l \times b \times h$ or $\pi r^2 h$. Use $\pi \approx 3.142$.

Step 4 — Add the correct unit

Length = m. Area = m^2 . Volume = m^3 or L. Round to 2 decimal places.

● **Tip:** $1\,m^3 = 1\,000\,L$. This conversion is tested often in water and tank questions.

WORKED EXAMPLE A

Easy

A rectangular garden is 8 m long and 5 m wide. Calculate the area.

Step 1 Length = 8 m, width = 5 m.

Step 2 Formula: Area = $l \times b$

Step 3 = 8×5

Step 4 = $40\,m^2$

✓ **Area = $40\,m^2$**

WORKED EXAMPLE B

Medium

A cylindrical water tank has radius 1.5 m and height 2 m. Calculate the volume in litres. Use $\pi = 3.142$.

Step 1 $r = 1.5$ m, $h = 2$ m.

Step 2 Formula: $V = \pi r^2 h$

Step 3 $V = 3.142 \times (1.5)^2 \times 2 = 3.142 \times 2.25 \times 2$

Step 4 $V = 14.14$ m³

Step 5 Convert: $14.14 \times 1\,000 = 14\,139$ L

✓ **Volume = 14 139 L**

WORKED EXAMPLE C

Hard

A room is 4 m by 3 m. Tiles are 30 cm by 30 cm. How many tiles are needed to cover the floor?

Step 1 Floor area = $4 \times 3 = 12$ m².

Step 2 Convert: 12 m² = $12 \times 10\,000 = 120\,000$ cm².

Step 3 Tile area = $30 \times 30 = 900$ cm².

Step 4 Tiles needed = $120\,000 \div 900 = 133.33$

Step 5 Round up: 134 tiles.

✓ **134 tiles needed**

Try It Yourself

A rectangular pool is 10 m long, 6 m wide, and 2 m deep. Calculate the volume in m³.

Answer: $V = l \times b \times h = 10 \times 6 \times 2 = 120$ m³

METHOD
3

Maps and Scale

Step 1 — Read the scale

Scale 1 : 100 means 1 cm on the map = 100 cm in real life.

Step 2 — Measure or find the map distance

Use a ruler or read the value given in the question.

Step 3 — Multiply or divide

Real distance = Map distance $\times$ scale. Map distance = Real $\div$ scale.

Step 4 — Convert units if needed

Change cm to m or m to km. Always check the unit asked for.

● **Tip:** Going from map to real life: **multiply**. Going from real life to map: **divide**.

WORKED EXAMPLE A

Easy

A map has a scale of 1 : 50. A line on the map is 4 cm. Calculate the real length in cm.

Step 1 Scale = 1 : 50.

Step 2 Map distance = 4 cm.

Step 3 Real = $4 \times 50 = 200$ cm

✓ **Real length = 200 cm (or 2 m)**

WORKED EXAMPLE B

Medium

A floor plan has a scale of 1 : 100. The lounge measures 5 cm by 3 cm on the plan. Find the real area of the lounge in m^2 .

Step 1 Real length = $5 \times 100 = 500$ cm = 5 m.

Step 2 Real width = $3 \times 100 = 300$ cm = 3 m.

Step 3 Area = $5 \times 3 = 15$ m^2

✓ **Real area = 15 m^2**

WORKED EXAMPLE C

Hard

A road map has a scale of 1 : 250 000. Two towns are 8 cm apart on the map. A car travels at 80 km/h. How long will the trip take?

Step 1 Real distance = $8 \times 250\,000 = 2\,000\,000$ cm.

Step 2 Convert: $2\,000\,000 \text{ cm} \div 100\,000 = 20$ km.

Step 3 Time = Distance $\div$ Speed = $20 \div 80$

Step 4 Time = 0.25 hours = 15 minutes

✓ **Trip time = 15 minutes**

Try It Yourself

A wall on a plan is 6 cm. The scale is 1 : 50. What is the real length in metres?

Answer: $6 \times 50 = 300$ cm = 3 m

METHOD
4

Data Handling and Graphs

Step 1 — Sort the values

Write numbers from smallest to largest. This helps with median, mode, and range.

Step 2 — Choose the right formula

Mean = Sum $\div$ count. Median = middle. Mode = most often. Range = max – min.

Step 3 — Read graphs carefully

Check axes, scale, and units. Find the correct row or column in tables.

Step 4 — Show the calculation

Write the formula and substitute. Don't just write the answer.

● **Tip:** If there are 2 middle numbers, the median is the average of the two.

WORKED EXAMPLE A

Easy

The marks of 5 learners are: 60, 75, 80, 50, 90. Calculate the mean.

Step 1 Sum = $60 + 75 + 80 + 50 + 90 = 355$.

Step 2 Formula: Mean = Sum $\div$ count.

Step 3 Mean = $355 \div 5 = 71$

✓ **Mean = 71**

WORKED EXAMPLE B

Medium

Find the median and range: 12, 8, 15, 22, 9, 18, 14.

Step 1 Sort: 8, 9, 12, 14, 15, 18, 22.

Step 2 Middle value (4th of 7) = 14.

Step 3 Range = $22 - 8 = 14$.

✓ **Median = 14 ; Range = 14**

WORKED EXAMPLE C

Hard

A class has 30 learners. 12 are boys. One learner is chosen at random. What is the probability that a girl is chosen? Give the answer as a percentage.

Step 1 Number of girls = $30 - 12 = 18$.

Step 2 Formula: P = favourable $\div$ total.

Step 3 P(girl) = $18 \div 30 = 0.6$

Step 4 As percentage: $0.6 \times 100 = 60\%$

✓ **P(girl) = 60%**

Try It Yourself

Find the mode of: 4, 7, 3, 7, 9, 4, 7, 2, 7.

Answer: Mode = 7 (it appears 4 times)

METHOD
5

Reading Graphs and Tables

Step 1 — Read the title and axes

What does each axis show? What are the units? Read every label.

Step 2 — Check the scale

One block may equal 5, 10, 50, or 100. Look closely at the gaps.

Step 3 — Find the value asked for

Use a ruler. Trace from the x-axis up to the bar or line. Then across.

Step 4 — Compare or calculate

For "highest" or "increase," compare two values. Show the subtraction.

● **Tip:** Percentage change = $(\text{New} - \text{Old}) \div \text{Old} \times 100$. Always divide by the OLD value.

WORKED EXAMPLE A

Easy

A bar graph shows sales: Jan = R200, Feb = R350, Mar = R500. What is the total for the 3 months?

Step 1 Read each bar: 200, 350, 500.

Step 2 Add: $200 + 350 + 500 = 1\ 050$

✓ **Total = R1 050**

WORKED EXAMPLE B

Medium

The price of bread was R12 in 2024. In 2025 it was R15. Calculate the percentage increase.

Step 1 Old = R12. New = R15.

Step 2 Formula: % change = $(\text{New} - \text{Old}) \div \text{Old} \times 100$

Step 3 = $(15 - 12) \div 12 \times 100$

Step 4 = $3 \div 12 \times 100 = 25\%$

✓ **Percentage increase = 25%**

WORKED EXAMPLE C

Hard

A water tariff is: First 6 kL free. Next 6 kL at R12/kL. Above 12 kL at R20/kL. A household uses 18 kL. Calculate the total bill.

Step 1 First 6 kL: free = R0.

Step 2 Next 6 kL (6-12): $6 \times R12 = R72$.

Step 3 Above 12 kL: $18 - 12 = 6$ kL. $6 \times R20 = R120$.

Step 4 Total = $0 + 72 + 120 = R192$

✓ **Total water bill = R192**

Try It Yourself

An item cost R250. The price is now R200. Find the percentage decrease.

Answer: $(250 - 200) \div 250 \times 100 = 20\% \text{ decrease}$

PART 4

Mistakes That Cost Students Marks

These are the most common reasons marks are dropped — even by students who understand the work.

1. Missing Units in the Final Answer

✗ Writing "Area = 40" instead of "Area = 40 m²". Loses a mark every time.

✓ Add the unit after every answer. R, %, m, m², m³, kg, L.

👉 Means the student forgot the unit was part of the answer.

2. Using the Wrong Formula

✗ Using simple interest when compound interest is needed. Or area instead of volume.

✓ Read the question first. Underline the words "compound," "volume," "area." Then pick the formula.

👉 Means the student rushed and didn't read the question properly.

3. Rounding Too Early

✗ Rounding 3.142 to 3 in the middle of a calculation. Final answer becomes wrong.

✓ Keep all decimals during the calculation. Round only at the very end.

👉 Means the student doesn't trust the calculator's full answer.

4. Forgetting VAT Included vs Excluded

✗ Multiplying by 1.15 when the price already has VAT. Adding VAT twice.

✓ "Excluding VAT" → multiply by 1.15. "Including VAT" → divide by 1.15. Read every word.

👉 Means the student didn't notice the key words in the question.

5. Mixing Units in the Same Calculation

✗ Adding 50 cm to 2 m and writing 52. The units are different.

✓ Convert both to the same unit first. 50 cm = 0.5 m, so 0.5 + 2 = 2.5 m.

👉 Means the student doesn't check units before calculating.

6. Misreading Graph Scales

✗ Reading "20" when each block is worth 5, so the value is actually 100.

✓ Always check what one block equals. Look at the axis labels first.

👉 Means the student didn't pause to study the graph before answering.

7. Skipping Working Out

✗ *Writing only the final answer. If it's wrong, you get zero. No method marks.*

✓ Show formula → substitution → calculation → answer. You earn marks for each step.

👉 Means the student doesn't know that method marks are worth more than the answer.

8. Wrong Calculator Order of Operations

✗ *Typing $5 + 3 \times 2$ and getting 16. The calculator gives 11 (BODMAS).*

✓ Use brackets in the calculator: $(5 + 3) \times 2 = 16$. Or do steps separately.

👉 Means the student trusts the calculator without checking the order.

PART 5

Your 14-Day Study Plan

Follow this plan day by day. Topics earlier in the plan are worth the most marks. Practice every day. Do not skip ahead.

Days 1-3

Percentages, VAT & Finance

~35
marks

- Practise 20 VAT calculations: add VAT and remove VAT.
- Do 15 percentage increase / decrease problems.
- Do 10 simple and 10 compound interest calculations.
- Practise 5 salary slip questions: gross, deductions, net.
- Goal: Solve any finance question in under 4 minutes.

Days 4-5

Tables and Tariffs

~20
marks

- Practise 10 tariff problems: water, electricity, phone bills.
- Read 10 tax tables. Find the correct bracket every time.
- Practise reading rows and columns under exam pressure.
- Goal: Find any value in a table in under 30 seconds.

Days 6-7

Measurement and Conversions

~25
marks

- Calculate 15 areas: rectangles, triangles, circles.
- Calculate 10 volumes: boxes, cylinders.
- Practise 20 unit conversions: mm, cm, m, km, mL, L, g, kg.
- Goal: No mistakes on units. No mistakes on formulas.

Days 8-9

Maps, Scale, and Plans

~15
marks

- Practise 10 scale calculations: 1 : 50, 1 : 100, 1 : 250 000.
- Read 5 floor plans. Calculate real lengths and areas.
- Practise 5 distance / time / speed problems.
- Goal: Move from map to real life without confusion.

Days 10-11

Data Handling and Graphs

~30
marks

- Practise 10 mean / median / mode / range problems.
- Read 10 different graphs: bar, line, pie, box-and-whisker.
- Practise 10 probability questions. Give answer as fraction, decimal, AND percentage.
- Goal: Full marks on Q3 every time.

Day 12**Verify and Justify Questions***~10
marks*

- Practise 10 "verify" questions. Show calculation, then answer "yes" or "no."
- Don't just write the answer — show the maths that proves it.
- Goal: Earn full marks by showing every step.

Day 13**Timed Mixed Practice***Test
yourself*

- Set a timer. Do Q1 (35 min). Q2 (35 min). Q3 (30 min).
- Work in silence. No notes. No phone.
- Mark your answers. Note where you lost marks.
- Redo only the parts you got wrong.

Day 14**Full Past Paper — Exam Simulation***Exam
mode*

- Print one full past Paper 1. Most recent year.
- Set a 3-hour timer. Silence your phone. No notes.
- When done, mark it. Write your total score.
- List 3 things to fix. Revise those before the real exam.

EVERY DAY**Daily 5-Minute Warm-Up**

Daily Task	Why
✓ 3 percentage calculations	Keeps VAT and discount work fast and automatic.
✓ 1 graph or table read	Trains your eye to find values quickly.
✓ 1 unit conversion	Stops you mixing cm, m, and km in the exam.
✓ 1 finance problem (interest)	Locks in the formula $A = P(1 + i)^n$.
✓ Say one formula out loud	Hearing yourself helps you remember.

**You can do this. The paper is predictable.
The methods are simple. Master the 5 core topics,
avoid the 8 common mistakes, and follow the 14-day plan.
That is all you need to do.**

PREDICTIVE 2026 MATHEMATICAL LITERACY PAPER 1

PAPER A — QUESTION PAPER

Time: 3 hours | **Total marks:** 150 | **Grade:** 12 | **Date:** 2026

READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This question paper consists of FIVE (5) questions. Answer ALL the questions.
2. Number the answers correctly according to the numbering system used in this paper.
3. Start each question on a NEW page.
4. Show ALL calculations, formulae and substitutions clearly.
5. Round off ALL final answers to TWO decimal places, unless stated otherwise.
6. ALL answers must include the correct units (R, %, m, m², m³, kg, L, etc.).
7. Use the recommended time allocation. Do NOT spend too much time on any one question.
8. Maps, plans and diagrams are NOT necessarily drawn to scale, unless stated.
9. You may use a non-programmable calculator.
10. Write neatly and legibly.

QUESTION	TOPIC	MARKS	TIME
1	Basic calculations & tables	30	35 min
2	Finance (income, VAT, tax, interest)	35	40 min
3	Data handling & probability	30	35 min
4	Measurement, tariffs & maps	30	35 min
5	Mixed application (currency, interest, % change)	25	35 min
TOTAL		150	180 min

Each question contains its own SOURCE MATERIAL (tables, graphs, diagrams or text). Read the information carefully before attempting the sub-questions.

QUESTION 1 — BASIC CALCULATIONS AND TABLES

Marks: 30 · Time: 35 min

INFORMATION 1 — TILL SLIP FROM A SUPERMARKET

Sarah went grocery shopping at her local supermarket. Her till slip is shown in TABLE 1 below. All prices include VAT (Value Added Tax).

ITEM	QUANTITY	UNIT PRICE (R)	TOTAL (R)
Brown bread	2	18,00	A
Milk (2 L)	1	32,99	32,99
Apples	5	6,50	32,50
Sugar (2,5 kg)	1	64,95	64,95
Cooking oil (750 mL)	1	89,99	89,99
Eggs (1 dozen)	1	49,50	49,50
SUB-TOTAL (incl. VAT)			B

[VAT in South Africa is currently 15%]

1.1 Refer to TABLE 1 above. Use it to answer the following questions.

1.1.1 Calculate the value of **A** on the till slip. (2)

1.1.2 Calculate the value of **B**, the sub-total of the bill (including VAT). (2)

1.1.3 Calculate the amount of VAT included in the sub-total **B**. Round your answer to TWO decimal places. (3)

1.1.4 Sarah pays the cashier with two R200 notes. Calculate her change. (2)

1.1.5 Write the cost of the brown bread (value **A**) to the cost of the eggs as a ratio in simplest form. (3)

Subtotal: (12)

INFORMATION 2 — A GRADE 12 CLASS

There are 30 learners in Sarah's Grade 12 class. The class consists of 18 girls and 12 boys. The teacher chooses ONE learner at random to answer a question.

1.2.1 Write the number of boys to the total number of learners as a ratio in simplest form. (2)

1.2.2 Determine the probability (as a decimal, rounded to TWO decimal places) that a girl is chosen. (2)

1.2.3 Write your answer to QUESTION 1.2.2 as a percentage. (1)

Subtotal: (5)**INFORMATION 3 — SIPHO'S MONTHLY CELLPHONE COSTS**

Sipho's cellphone usage and the rates charged by his service provider are shown in TABLE 2 below.

SERVICE	USAGE	RATE
Voice calls	240 minutes per month	R1,20 per minute
SMS messages	80 messages per month	R0,50 per message
Mobile data	2 GB per month	R49,00 per GB

- 1.3.1** Calculate Sipho's TOTAL monthly cellphone cost. **(4)**
- 1.3.2** The next month, the service provider increased ALL rates by 8%. Calculate the new total monthly cost. Round to TWO decimal places. **(3)**
- 1.3.3** Express the cost of voice calls as a percentage of the original total cost calculated in QUESTION 1.3.1. Round to TWO decimal places. **(3)**
- 1.3.4** Convert 240 minutes into hours and minutes. **(3)**

Subtotal: (13)

TOTAL FOR QUESTION 1: [30]

QUESTION 2 — FINANCE

Marks: 35 · Time: 40 min

INFORMATION 4 — MR PATEL'S MONTHLY PAYSリップ

Mr Patel works as an accountant. His monthly payslip for May 2026 is shown in TABLE 3 below.

EARNINGS (R)		DEDUCTIONS (R)	
Basic salary	28 500,00	PAYE Tax	4 200,00
Travel allowance	2 500,00	UIF (1% of basic salary)	285,00
Performance bonus	1 500,00	Medical aid	1 850,00
		Pension fund	1 425,00
GROSS SALARY	A	TOTAL DEDUCTIONS	B
NET SALARY			C

- 2.1.1** Calculate the gross salary **A**. (2)
- 2.1.2** Calculate the total deductions **B**. (2)
- 2.1.3** Calculate the net salary **C**. (2)
- 2.1.4** Calculate the PAYE Tax as a percentage of the gross salary. Round to TWO decimal places. (3)
- 2.1.5** UIF is 1% of the basic salary. Show by calculation whether the UIF amount on the payslip is correct. (3)

Subtotal: (12)

INFORMATION 5 — TV PURCHASE

Mr Patel sees a smart TV advertised for **R8 999,00** (VAT included).

- 2.2.1** Calculate the price of the TV BEFORE VAT was added. Round to TWO decimal places. (3)
- 2.2.2** The shop offers a 10% discount on the advertised price. Calculate the new price (including VAT) after the discount. (3)
- 2.2.3** Determine how much money Mr Patel saved by getting the discount. (2)

Subtotal: (8)

INFORMATION 6 — INCOME TAX TABLE (SIMPLIFIED)

The simplified annual income tax brackets for the 2026 tax year are shown in TABLE 4 below.

BRACKET	TAXABLE INCOME (PER YEAR)	RATE OF TAX
1	R 0 – R 237 100	18% of taxable income
2	R 237 101 – R 370 500	R 42 678 + 26% of amount above R 237 100
3	R 370 501 – R 512 800	R 77 362 + 31% of amount above R 370 500

Mr Patel's annual taxable income is **R 385 000**.

- 2.3.1** Show by calculation in which tax bracket Mr Patel falls. **(2)**
- 2.3.2** Calculate Mr Patel's annual tax according to the tax table. Show ALL calculations. **(5)**
- 2.3.3** Calculate his MONTHLY tax payment based on your answer to QUESTION 2.3.2. **(2)**
Round to TWO decimal places.

Subtotal: (9)

INFORMATION 7 — INVESTMENT OPTIONS

Mr Patel wants to invest **R 20 000** for **3 years**. The bank offers him two options:

- **Option A:** 9% simple interest per year
- **Option B:** 8,5% compound interest per year (compounded annually)

- 2.4.1** Calculate the value of his investment under Option A after 3 years. **(3)**
Use the formula: **$A = P(1 + i \times n)$**
- 2.4.2** Calculate the value of his investment under Option B after 3 years. **(3)**
Use the formula: **$A = P(1 + i)^n$** . Round to TWO decimal places.

Subtotal: (6)

TOTAL FOR QUESTION 2: [35]

QUESTION 3 — DATA HANDLING AND PROBABILITY

Marks: 30 · Time: 35 min

INFORMATION 8 — SCREEN-TIME SURVEY

The daily screen-time (in hours) of 30 Grade 12 learners was recorded. The data is shown below, sorted from lowest to highest:

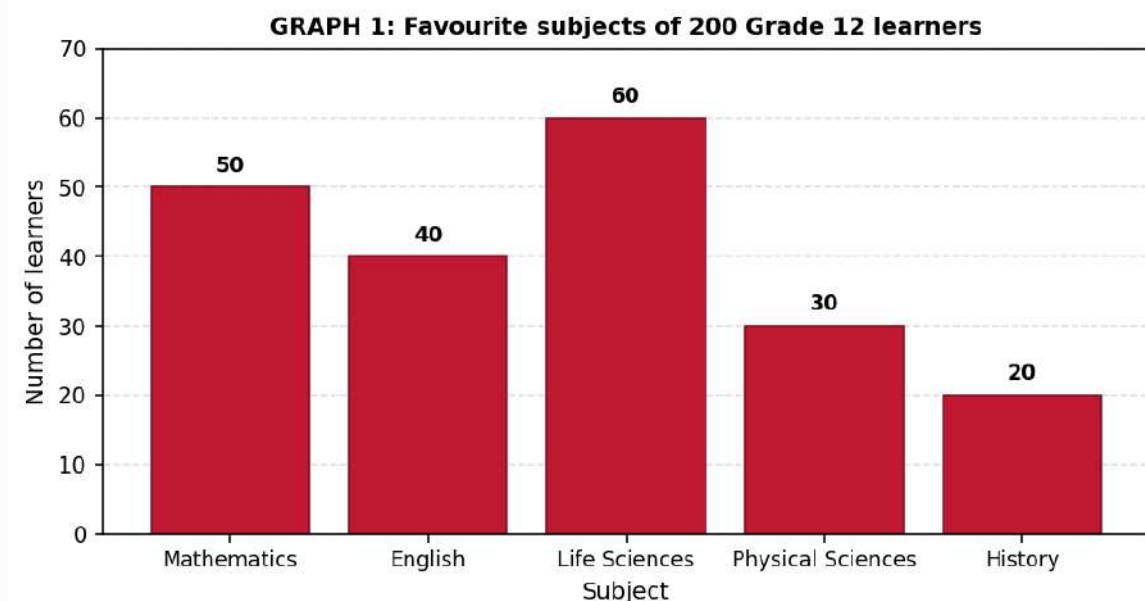
1,5	2,0	2,5	3,0	3,0	3,5	4,0	4,0	4,5	5,0	5,0	5,0	5,5
6,0	6,0	6,0										
6,5	6,5	7,0	7,0	7,5	8,0	8,0	8,5	9,0	9,0	9,5	10,0	10,5
11,0												

- 3.1.1** Determine the range of the data. (2)
- 3.1.2** Calculate the mean (average) screen-time. Round to TWO decimal places. (3)
- 3.1.3** Determine the median of the data. (2)
- 3.1.4** Determine the mode of the data. (2)
- 3.1.5** A learner is chosen at random from the survey. Calculate the probability that the learner spends MORE THAN 8 hours per day on a screen. Express your answer as a percentage. Round to TWO decimal places. (3)

Subtotal: (12)

INFORMATION 9 — FAVOURITE SUBJECT BAR GRAPH

The bar graph below shows the favourite school subjects of 200 Grade 12 learners.



- 3.2.1** Which subject is the MOST popular? (1)
- 3.2.2** How many learners chose Mathematics OR English as their favourite subject? (2)
- 3.2.3** Express the number of learners who chose Life Sciences as a percentage of the total. Round to TWO decimal places. (3)
- 3.2.4** A learner is chosen at random. Calculate the probability that this learner chose History as a favourite subject. Give your answer as a fraction in simplest form. (3)

Subtotal: (9)

INFORMATION 10 — KHUMALO FAMILY MONTHLY BUDGET

The Khumalo family has a monthly income of **R 18 500**. The percentages of income spent on different categories are shown in TABLE 5 below.

CATEGORY	PERCENTAGE OF INCOME
Rent	35%
Food	25%
Transport	15%
Savings	10%
Other expenses	15%
TOTAL	100%

- 3.3.1** Calculate the amount the family spends on FOOD per month. (2)
- 3.3.2** Calculate how much money the family saves in ONE YEAR. (3)
- 3.3.3** Mr Khumalo claims: "*We spend more on rent than on food and savings combined.*" (4)
Verify, with calculations, whether his statement is CORRECT or INCORRECT.

Subtotal: (9)

TOTAL FOR QUESTION 3: [30]

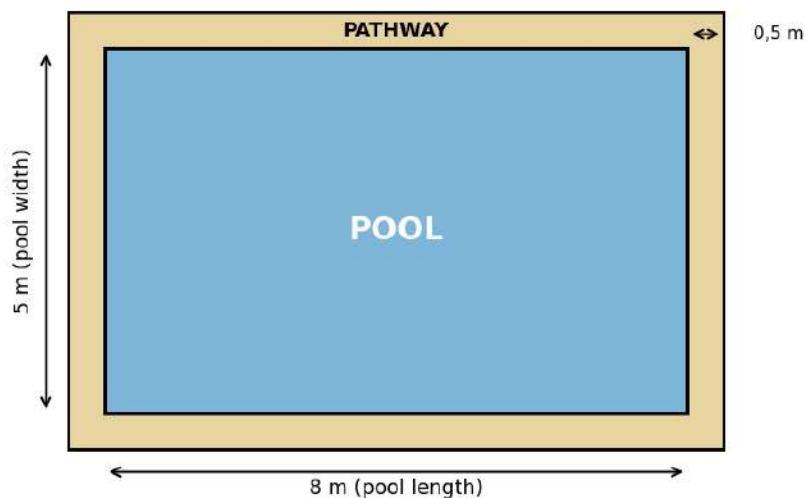
QUESTION 4 — MEASUREMENT, TARIFFS AND MAPS

Marks: 30 · Time: 35 min

INFORMATION 11 — THE NAIDOO FAMILY SWIMMING POOL

The Naidoo family has a rectangular swimming pool in their backyard. The pool measures **8 m long** and **5 m wide**. It is filled to a uniform depth of **1,5 m**. They plan to build a paved pathway of width **0,5 m** all the way around the outside of the pool.

DIAGRAM A: Top view of swimming pool (depth = 1,5 m) with pathway



Useful formulas:

Area of rectangle = length $\times$ width | Volume of rectangular prism = length $\times$ width $\times$ height
 $1 \text{ m}^3 = 1\,000 \text{ L}$

- 4.1.1** Calculate the area of the WATER SURFACE of the pool (in m^2). (2)
- 4.1.2** Calculate the volume of water in the pool, in cubic metres (m^3). (2)
- 4.1.3** Convert your answer in QUESTION 4.1.2 to litres (L). (2)
- 4.1.4** Calculate the area of the PATHWAY only (the shaded outer ring around the pool). (5)
 Show ALL calculations.
 (Hint: Area of pathway = Area of outer rectangle – Area of pool)

Subtotal: (11)

INFORMATION 12 — MUNICIPAL WATER TARIFF

The municipality charges water on a STEP TARIFF (block tariff) according to TABLE 6 below.

BLOCK	VOLUME PER MONTH (kL)	COST PER kL (R)
1	0 – 6	FREE

BLOCK	VOLUME PER MONTH (kL)	COST PER kL (R)
2	more than 6, up to 15	13,80
3	more than 15, up to 30	22,50
4	more than 30	35,00

The Naidoo family used **24 kL** of water during the month of May.

- 4.2.1** Calculate the family's TOTAL water bill for May. Show ALL calculations. (5)
- 4.2.2** Calculate the average cost per kL of water used by the family. Round to TWO decimal places. (3)
- 4.2.3** Suggest ONE way the family could reduce their water bill. (1)

Subtotal: (9)

INFORMATION 13 — ROAD TRIP FROM DURBAN TO JOHANNESBURG

The Naidoo family is planning a road trip from Durban to Johannesburg. On their map, the scale is **1 : 5 000 000**. The straight-line distance between the two cities measured on the map is **11,4 cm**.

Note: 1 km = 100 000 cm

- 4.3.1** Calculate the actual (real) distance between Durban and Johannesburg, in kilometres (km). (3)
- 4.3.2** If they travel at an average speed of 100 km/h, calculate how long the trip will take. Give your answer in hours and minutes. (4)
- 4.3.3** The family's car uses 8 L of petrol per 100 km. Calculate how many litres of petrol they will use for the trip. (3)

Subtotal: (10)

TOTAL FOR QUESTION 4: [30]

QUESTION 5 — MIXED APPLICATION (CURRENCY, INTEREST, % CHANGE)

Marks: 25 · Time: 35 min

INFORMATION 14 — MRS MTHEMBU'S TRIP TO LONDON

Mrs Mthembu plans to visit her son who lives in London, United Kingdom. The exchange rate at the time of her trip is:

$$£ 1 = R 23,80$$

(£ = British pound | R = South African rand)

- 5.1.1** Mrs Mthembu has saved R 28 560,00 for her trip. Convert this amount to British pounds (£). **(2)**
- 5.1.2** In London, she buys a winter coat for £ 75,00. Calculate the cost in South African rand (R). **(2)**
- 5.1.3** Her flight ticket costs R 12 500,00. The travel agent offers a 5% early-booking discount. Calculate the discounted price of the flight ticket. **(3)**

Subtotal: (7)

INFORMATION 15 — INVESTMENT COMPARISON

Mrs Mthembu wants to invest **R 50 000** for **4 years**. She compares two banks:

- **Bank A:** 7% simple interest per year
- **Bank B:** 6,5% compound interest per year (compounded annually)

Use: Simple interest: $A = P(1 + i \times n)$ Compound interest: $A = P(1 + i)^n$

- 5.2.1** Calculate the final amount of her investment under **Bank A** after 4 years. **(3)**
- 5.2.2** Calculate the final amount of her investment under **Bank B** after 4 years. Round to TWO decimal places. **(4)**
- 5.2.3** Determine which bank gives the BETTER investment. Justify your answer using the calculations above. **(3)**

Subtotal: (10)

INFORMATION 16 — SALARY INCREASE vs INFLATION

Mrs Mthembu earned **R 285 000,00** in 2024 and **R 310 000,00** in 2025. The inflation rate for 2025 was **6,5%**.

- 5.3.1** Calculate the percentage increase in her annual earnings from 2024 to 2025. **(3)**
Round to TWO decimal places.
- 5.3.2** VERIFY, by calculation, whether her salary increase kept up with inflation. Show **(5)**
ALL working AND justify your final answer.

Subtotal: (8)

TOTAL FOR QUESTION 5: [25]

GRAND TOTAL: 150 MARKS

— END OF QUESTION PAPER —

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PREDICTIVE 2026 MATHEMATICAL LITERACY PAPER 1

PAPER A — MEMORANDUM

Marking notes: Accept alternative correct methods. Award method marks even if a final answer is incorrect (consequential marks). Penalise the omission of units only ONCE per question. Round only at the FINAL step unless instructed otherwise.

Question	Q1	Q2	Q3	Q4	Q5	TOTAL
Marks	30	35	30	30	25	150

QUESTION 1 — MEMORANDUM [30 marks]

1.1 Sarah's Till Slip

1.1.1 Calculate value A (cost of brown bread).

$$A = \text{quantity} \times \text{unit price}$$

$$A = 2 \times \text{R}18,00$$

$$\checkmark A = \text{R}36,00$$

(2)

1.1.2 Calculate value B (sub-total including VAT).

$$B = 36,00 + 32,99 + 32,50 + 64,95 + 89,99 + 49,50$$

$$\checkmark B = \text{R}305,93$$

(2)

1.1.3 Calculate the VAT included in B.

$$\text{VAT included} = B \times 15 \div 115$$

$$\text{VAT} = 305,93 \times 15 \div 115$$

$$\text{VAT} = 4\,588,95 \div 115$$

$$\checkmark \text{VAT} = \text{R}39,90$$

(3)

1.1.4 Sarah pays with two R200 notes. Calculate her change.

$$\text{Change} = (2 \times \text{R}200) - \text{R}305,93 = \text{R}400 - \text{R}305,93$$

$$\checkmark \text{Change} = \text{R}94,07$$

(2)

1.1.5 Write the cost of bread to the cost of eggs as a ratio.

Bread : Eggs = R36,00 : R49,50

Multiply both by 2 to remove decimals: 72 : 99

Divide both by 9 (HCF): 8 : 11

✓ **Ratio = 8 : 11**

(3)

1.2 Class Probability

1.2.1 Boys to total ratio.

Boys : Total = 12 : 30

Divide both by 6 (HCF): 2 : 5

✓ **Ratio = 2 : 5**

(2)

1.2.2 Probability that a girl is chosen (decimal).

$P(\text{girl}) = \text{number of girls} \div \text{total} = 18 \div 30 = 0,6$

✓ **P(girl) = 0,60**

(2)

1.2.3 Express as a percentage.

$0,60 \times 100 = 60$

✓ **P(girl) = 60%**

(1)

1.3 Sipho's Cellphone Costs

1.3.1 Total monthly cellphone cost.

Calls = $240 \times R1,20 = R288,00$

SMS = $80 \times R0,50 = R40,00$

Data = $2 \times R49,00 = R98,00$

Total = $R288,00 + R40,00 + R98,00$

✓ **Total = R426,00**

(4)

1.3.2 New total after 8% increase.

New total = old total $\times 1,08$

New total = $R426,00 \times 1,08 = R460,08$

✓ **New total = R460,08**

(3)

1.3.3 Calls as % of total in 1.3.1.

$\% = (\text{calls} \div \text{total}) \times 100 = (288 \div 426) \times 100$

$$\% = 0,67605... \times 100 = 67,605...$$

$$\checkmark \% = \mathbf{67,61\%}$$

(3)

1.3.4 Convert 240 minutes to hours and minutes.

$$240 \div 60 = 4 \text{ hours exactly (since } 60 \times 4 = 240)$$

Remaining minutes = 0

$$\checkmark \mathbf{240 \text{ minutes} = 4 \text{ hours } 0 \text{ minutes}}$$

(3)

QUESTION 2 — MEMORANDUM [35 marks]**2.1 Mr Patel's Payslip**

2.1.1 Calculate gross salary A.

$$A = \text{Basic} + \text{Travel} + \text{Bonus} = 28\,500 + 2\,500 + 1\,500$$

✓ **A = R32 500,00**

(2)

2.1.2 Calculate total deductions B.

$$B = \text{PAYE} + \text{UIF} + \text{Medical} + \text{Pension}$$

$$B = 4\,200 + 285 + 1\,850 + 1\,425$$

✓ **B = R7 760,00**

(2)

2.1.3 Calculate net salary C.

$$C = \text{Gross} - \text{Total deductions} = 32\,500 - 7\,760$$

✓ **C = R24 740,00**

(2)

2.1.4 PAYE as % of gross salary.

$$\% = (\text{PAYE} \div \text{Gross}) \times 100 = (4\,200 \div 32\,500) \times 100$$

$$\% = 0,12923... \times 100 = 12,9230...$$

✓ **% = 12,92%**

(3)

2.1.5 Verify UIF (1% of basic).

$$\text{UIF} = 1\% \times \text{R}28\,500 = 0,01 \times 28\,500 = \text{R}285,00$$

The amount on the payslip is R285,00.

✓ **The UIF amount is CORRECT.**

(3)

2.2 TV Purchase

2.2.1 Price BEFORE VAT.

$$\text{Price before VAT} = \text{Price (incl)} \div 1,15$$

$$= \text{R}8\,999,00 \div 1,15 = 7\,825,217...$$

✓ **Price before VAT = R7 825,22**

(3)

2.2.2 New price after 10% discount.

$$\text{Discounted price} = \text{R}8\,999,00 \times (1 - 0,10) = \text{R}8\,999,00 \times 0,90$$

✓ **Discounted price = R8 099,10**

(3)

2.2.3 Money saved with discount.

$$\text{Saving} = \text{R}8\,999,00 - \text{R}8\,099,10 = \text{R}899,90$$

$$(\text{OR: } 10\% \times \text{R}8\,999,00 = \text{R}899,90)$$

$$\checkmark \text{ Saving} = \text{R}899,90$$

(2)**2.3 Income Tax**

2.3.1 Identify the tax bracket.

R385 000 is between R370 501 and R512 800.

✓ **Mr Patel falls in BRACKET 3.**

(2)

2.3.2 Calculate annual tax.

$$\text{Tax} = \text{R}77\,362 + 31\% \times (385\,000 - 370\,500)$$

$$= \text{R}77\,362 + 0,31 \times 14\,500$$

$$= \text{R}77\,362 + \text{R}4\,495,00$$

$$\checkmark \text{ Annual tax} = \text{R}81\,857,00$$

(5)

2.3.3 Monthly tax.

$$\text{Monthly tax} = \text{R}81\,857,00 \div 12 = 6\,821,4166\dots$$

$$\checkmark \text{ Monthly tax} = \text{R}6\,821,42$$

(2)**2.4 Investment Options**

2.4.1 Option A (simple interest).

$$A = P(1 + i \times n) = 20\,000 \times (1 + 0,09 \times 3)$$

$$= 20\,000 \times (1 + 0,27) = 20\,000 \times 1,27$$

$$\checkmark \text{ A} = \text{R}25\,400,00$$

(3)

2.4.2 Option B (compound interest).

$$A = P(1 + i)^n = 20\,000 \times (1 + 0,085)^3$$

$$= 20\,000 \times (1,085)^3$$

$$= 20\,000 \times 1,277289\dots$$

$$\checkmark \text{ A} = \text{R}25\,545,78$$

(3)

QUESTION 3 — MEMORANDUM [30 marks]

3.1 Screen-Time Survey

3.1.1 Range.

Range = Highest – Lowest = 11,0 – 1,5

✓ **Range = 9,5 hours**

(2)

(2)

3.1.2 Mean (average).

Sum =
1,5+2,0+2,5+3,0+3,0+3,5+4,0+4,0+4,5+5,0+5,0+5,0+5,5+6,0+6,0+6,0+6,5+6,5+7,0+7,0+7,5+
Sum = 184,0
Mean = Sum ÷ count = 184,0 ÷ 30 = 6,1333...

✓ **Mean = 6,13 hours**

(3)

Sum = 184,0

✓ Mean = 6,13 hours

(3)

3.1.3 Median.

30 values $\rightarrow$ median = average of 15th and 16th values

15th = 6,0 ; 16th = 6,0

Median = $(6,0 + 6,0) \div 2 = 6,0$

✓ **Median = 6,0 hours**

(2)

$$\text{Median} = (6,0 + 6,0) \div 2 = 6,0$$

(2)

3.1.4 Mode.

Looking at frequencies: 5,0 appears 3 times; 6,0 appears 3 times.

✓ **Mode = 5,0 hours AND 6,0 hours (data is bimodal)**

(2)

(2)

3.1.5 $P(\text{more than 8 hours})$ as %.

Values more than 8 hours: 8,5 ; 9,0 ; 9,0 ; 9,5 ; 10,0 ; 10,5 ; 11,0 → 7 values

$P = 7 \div 30 = 0,2333\dots$

As % = $0,2333 \times 100 = 23,33\dots$

✓ **P = 23,33%**

(3)

$$\text{As \%} = 0,2333 \times 100 = 23,33\ldots$$

(3)

3.2 Bar Graph

3.2.1 Most popular subject.

✓ **Life Sciences (60 learners)**
(1)

(1)

3.2.2 Maths OR English.

$$\text{Total} = 50 + 40 = 90$$

✓ **90 learners**

(2)

3.2.3 Life Sciences as % of total.

$$\% = (60 \div 200) \times 100 = 30,00$$

✓ **% = 30,00%**

(3)

3.2.4 P(History) as a fraction in simplest form.

$$P(\text{History}) = 20 \div 200 = 20/200$$

Simplify: divide top and bottom by 20

✓ **P(History) = 1/10**

(3)

3.3 Khumalo Family Budget

3.3.1 Amount on Food per month.

$$\text{Food} = 25\% \times \text{R}18\,500 = 0,25 \times 18\,500$$

✓ **Food = R4 625,00**

(2)

3.3.2 Savings per year.

$$\text{Monthly savings} = 10\% \times \text{R}18\,500 = \text{R}1\,850,00$$

$$\text{Yearly savings} = \text{R}1\,850,00 \times 12$$

✓ **Yearly savings = R22 200,00**

(3)

3.3.3 Verify Mr Khumalo's claim.

$$\text{Rent} = 35\%$$

$$\text{Food} + \text{Savings} = 25\% + 10\% = 35\%$$

35% (Rent) = 35% (Food + Savings) — they are EQUAL, not greater.

✓ **The statement is INCORRECT. Rent is equal to Food + Savings combined, not more.**

(4)

QUESTION 4 — MEMORANDUM [30 marks]**4.1 Swimming Pool**

4.1.1 Surface area of pool.

$$\text{Area} = \text{length} \times \text{width} = 8 \times 5$$

$$\checkmark \text{ Area} = 40 \text{ m}^2$$

(2)

4.1.2 Volume of water (m^3).

$$\text{Volume} = \text{length} \times \text{width} \times \text{depth} = 8 \times 5 \times 1,5$$

$$\checkmark \text{ Volume} = 60 \text{ m}^3$$

(2)

4.1.3 Volume in litres.

$$60 \text{ m}^3 \times 1\,000 \text{ L/m}^3 = 60\,000 \text{ L}$$

$$\checkmark \text{ Volume} = 60\,000 \text{ L}$$

(2)

4.1.4 Area of pathway.

$$\text{Outer length} = 8 + 0,5 + 0,5 = 9 \text{ m}$$

$$\text{Outer width} = 5 + 0,5 + 0,5 = 6 \text{ m}$$

$$\text{Outer area} = 9 \times 6 = 54 \text{ m}^2$$

$$\text{Pathway area} = \text{Outer area} - \text{Pool area} = 54 - 40$$

$$\checkmark \text{ Pathway area} = 14 \text{ m}^2$$

(5)

4.2 Water Tariff

4.2.1 Total water bill (24 kL).

$$\text{Block 1 (0–6 kL): } 6 \times \text{R}0,00 = \text{R}0,00$$

$$\text{Block 2 (6–15 kL): } 9 \times \text{R}13,80 = \text{R}124,20$$

$$\text{Block 3 (15–30 kL): } 24 - 15 = 9 \text{ kL} \rightarrow 9 \times \text{R}22,50 = \text{R}202,50$$

$$\text{Total} = 0 + 124,20 + 202,50$$

$$\checkmark \text{ Total bill} = \text{R}326,70$$

(5)

4.2.2 Average cost per kL.

$$\text{Average} = \text{Total bill} \div \text{Total kL} = 326,70 \div 24$$

$$= 13,6125$$

$$\checkmark \text{ Average} = \text{R}13,61 \text{ per kL}$$

(3)**4.2.3 Suggestion to reduce water bill.**

ANY ONE: Use less water (e.g. shorter showers); fix leaking taps; install low-flow showerheads; use a bucket instead of a hose; collect rainwater for the garden.

✓ **Any reasonable answer (1 mark)**

(1)

4.3 Road Trip Distance

4.3.1 Real distance (km).

Real distance = Map distance $\times$ scale = $11,4 \times 5\,000\,000$ cm
= 57 000 000 cm

Convert to km: $57\,000\,000 \div 100\,000 = 570$

✓ **Real distance = 570 km**

(3)**4.3.2 Travel time at 100 km/h.**

Time = Distance $\div$ Speed = $570 \div 100 = 5,7$ hours

5,7 hours = 5 hours + $0,7 \times 60$ minutes = 5 hours + 42 minutes

✓ **Travel time = 5 hours 42 minutes**

(4)**4.3.3 Petrol used.**

Petrol = $(570 \div 100) \times 8 = 5,7 \times 8$

✓ **Petrol used = 45,60 L**

(3)

QUESTION 5 — MEMORANDUM [25 marks]**5.1 Currency Conversion**

5.1.1 Convert R28 560 to £.

$$£ = R \div \text{exchange rate} = 28\,560 \div 23,80$$

✓ **£ 1 200,00**

(2)

5.1.2 Convert £75 to R.

$$R = £ \times \text{exchange rate} = 75 \times 23,80$$

✓ **R1 785,00**

(2)

5.1.3 Discounted flight price (5% off R12 500).

$$\text{Discount} = 5\% \times R12\,500 = 0,05 \times 12\,500 = R625,00$$

$$\text{Discounted price} = R12\,500 - R625 = R11\,875,00$$

$$(\text{OR: } R12\,500 \times 0,95 = R11\,875,00)$$

✓ **Discounted price = R11 875,00**

(3)

5.2 Investment Comparison

5.2.1 Bank A — simple interest, 7%, 4 years.

$$A = P(1 + i \times n) = 50\,000 \times (1 + 0,07 \times 4)$$

$$= 50\,000 \times (1 + 0,28) = 50\,000 \times 1,28$$

✓ **A = R64 000,00**

(3)

5.2.2 Bank B — compound interest, 6,5%, 4 years.

$$A = P(1 + i)^n = 50\,000 \times (1 + 0,065)^4$$

$$= 50\,000 \times (1,065)^4$$

$$= 50\,000 \times 1,286466\dots$$

✓ **A = R64 323,31**

(4)

5.2.3 Better bank — justify.

Bank A: R64 000,00

Bank B: R64 323,31

Difference: R64 323,31 – R64 000,00 = R323,31 more in Bank B

✓ **BANK B is the better option because it gives R323,31 more after 4 years.**

(3)

5.3 Salary Increase vs Inflation

5.3.1 % increase in earnings.

$$\begin{aligned}\% \text{ increase} &= (\text{New} - \text{Old}) \div \text{Old} \times 100 \\ &= (310\,000 - 285\,000) \div 285\,000 \times 100 \\ &= 25\,000 \div 285\,000 \times 100 = 8,7719...\end{aligned}$$

✓ **% increase = 8,77%**

(3)

5.3.2 Verify whether salary increase kept up with inflation.

Salary increase = 8,77% (from 5.3.1)

Inflation rate = 6,5%

Compare: 8,77% > 6,5%

Difference: 8,77% – 6,5% = 2,27 percentage points above inflation

✓ **YES, her salary increase (8,77%) DID keep up with inflation (6,5%) and was higher by 2,27 percentage points. Her real income improved.**

(5)

END OF MEMORANDUM — TOTAL: 150 MARKS

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